

FT HOOD AAF, TX, USA

WMO: 722570

Lat: 31.133N

Lon: 97.717W

Elev: 924

StdP: 14.21

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 03933

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	26.3	30.0	0.6	5.9	40.2	11.4	10.2	41.6	27.5	49.8	24.9	49.7	8.8	340	0.473

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	21.0	102.1	74.9	99.6	74.7	97.4	74.4	80.8	87.5	79.3	87.6	77.8	89.4	8.7	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
79.3	157.4	83.5	77.2	146.4	83.2	75.2	136.6	82.3	45.2	87.4	43.5	86.3	41.9	90.1	84.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
23.3	20.0	18.1	20.9	104.1	3.3	3.4	18.6	106.6	16.7	108.6	14.8	110.5	12.4	113.0		
			DB													
			WB													
				17.0	80.2	4.3	1.9	13.9	81.5	11.4	82.6	9.0	83.7	5.9	85.1	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	68.9	49.0	54.2	61.2	68.2	75.2	83.4	85.5	86.7	80.7	70.1	60.2	51.5	(6)
		DBStd	15.17	9.86	11.29	9.54	7.49	6.16	4.10	3.50	4.07	5.37	8.37	9.24	9.64	
		HDD50	369	139	79	26	2	0	0	0	0	2	22	99		
		HDD65	1749	503	327	183	53	8	0	0	0	1	49	195	430	
		CDD50	7267	108	197	373	546	782	1002	1100	1137	920	627	329	146	
		CDD65	3171	7	25	64	148	325	552	635	672	470	209	52	12	
		CDH74	36715	56	197	452	1220	2987	6797	8318	9172	5224	1886	334	72	
		CDH80	18639	5	46	94	352	1171	3571	4654	5366	2602	708	63	7	
(14)	Wind	WSAvg	8.7	8.7	9.7	10.2	10.5	9.5	9.0	8.3	7.4	6.9	7.7	8.1	8.1	(14)
(15)	Precipitation	PrecAvg	35.1	2.4	2.5	3.1	2.8	4.4	3.6	2.1	2.3	3.2	3.7	2.9	2.6	(15)
(16)		PrecMax	47.7	5.6	7.4	9.1	5.9	10.7	9.7	6.1	8.0	8.4	11.0	10.1	8.9	(16)
(17)		PrecMin	21.7	0.3	0.1	0.0	0.5	1.1	0.5	0.0	0.1	0.4	0.4	0.3	0.2	(17)
(18)		PrecStd	7.9	1.5	1.8	1.9	1.6	2.3	2.2	1.6	2.3	2.1	2.7	2.5	2.2	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	80.4	86.5	87.1	92.1	95.8	102.8	103.8	106.3	102.1	93.3	85.3	80.5		(19)
(20)		0.4% MCWB	61.1	60.3	64.8	70.2	74.9	74.9	75.1	76.1	73.3	70.7	68.7	64.2		(20)
(21)		2% DB	74.1	80.2	82.4	87.4	92.4	99.0	101.1	102.9	97.7	89.7	81.7	75.6		(21)
(22)		2% MCWB	58.4	58.9	65.3	68.8	73.5	75.3	74.7	75.7	72.9	70.1	67.0	62.9		(22)
(23)		5% DB	69.6	75.4	79.0	84.0	89.4	96.2	98.6	100.0	94.6	87.2	77.9	71.5		(23)
(24)		5% MCWB	56.7	61.1	64.5	68.3	72.7	75.3	74.4	74.9	72.8	69.3	65.3	60.8		(24)
(25)		10% DB	65.6	71.3	75.8	81.1	86.6	94.0	96.2	97.8	92.0	84.3	74.3	67.7		(25)
(26)		10% MCWB	54.4	59.5	62.9	66.9	72.0	75.1	74.4	74.7	72.6	68.2	63.6	58.7		(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4% WB	67.1	68.5	70.8	75.3	79.6	82.4	82.8	80.0	78.4	79.6	72.5	69.1		(27)
(28)		0.4% MCDB	72.9	75.3	79.7	84.4	86.1	88.8	89.1	90.3	89.7	83.1	78.6	73.0		(28)
(29)		2% WB	64.5	66.5	69.1	73.4	77.4	80.7	81.1	78.3	77.1	74.9	70.7	67.1		(29)
(30)		2% MCDB	70.5	72.1	76.0	82.0	85.2	87.8	87.3	91.2	88.1	82.8	76.8	72.7		(30)
(31)		5% WB	61.0	64.6	67.8	71.9	76.1	79.2	79.3	77.4	76.1	73.4	68.9	64.4		(31)
(32)		5% MCDB	66.9	70.6	74.5	79.6	84.3	87.7	86.8	92.0	86.8	81.9	74.7	69.1		(32)
(33)		10% WB	55.4	62.1	66.3	70.4	74.9	77.4	77.5	76.7	75.3	72.1	66.7	61.1		(33)
(34)		10% MCDB	62.8	69.5	72.9	77.9	83.2	88.5	90.5	90.9	85.7	79.9	72.8	66.0		(34)
(35)	Mean Daily Temperature Range	MDBR	20.7	21.1	20.9	20.7	18.5	19.6	20.3	21.0	20.3	21.2	20.3	20.0		(35)
(36)		5% DB MCDBR	27.4	25.0	23.4	22.6	21.4	22.2	22.9	23.4	23.1	23.0	22.1	24.6		(36)
(37)		5% WB MCWBR	15.9	13.8	10.9	10.8	9.2	6.1	5.1	5.6	6.7	9.5	11.7	15.0		(37)
(38)		5% WB MCDBR	21.9	20.2	18.9	19.1	18.1	19.7	19.5	21.1	18.9	18.4	18.0	20.4		(38)
(39)	Clear-Sky Solar Irradiance	taub	0.313	0.327	0.353	0.393	0.427	0.423	0.435	0.426	0.396	0.367	0.342	0.317		(40)
(41)		taud	2.519	2.482	2.423	2.333	2.294	2.345	2.346	2.356	2.388	2.453	2.468	2.511		(41)
(42)		Ebn at Noon	292	297	295	285	275	274	270	272	277	279	278	284		(42)
(43)		Edh at Noon	27	31	36	40	42	40	40	39	36	32	29	26		(43)
(44)	All-Sky Solar Radiation	RadAvg	956	1116	1446	1739	1844	2106	2137	1989	1656	1333	1007	844		(44)
(45)		RadStd	101	187	128	159	169	162	163	115	143	196	122	114		(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (45 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+128

Nomenclature: See separate page